Employee Management System

1. Database Definition

The database is designed to store and manage information related to

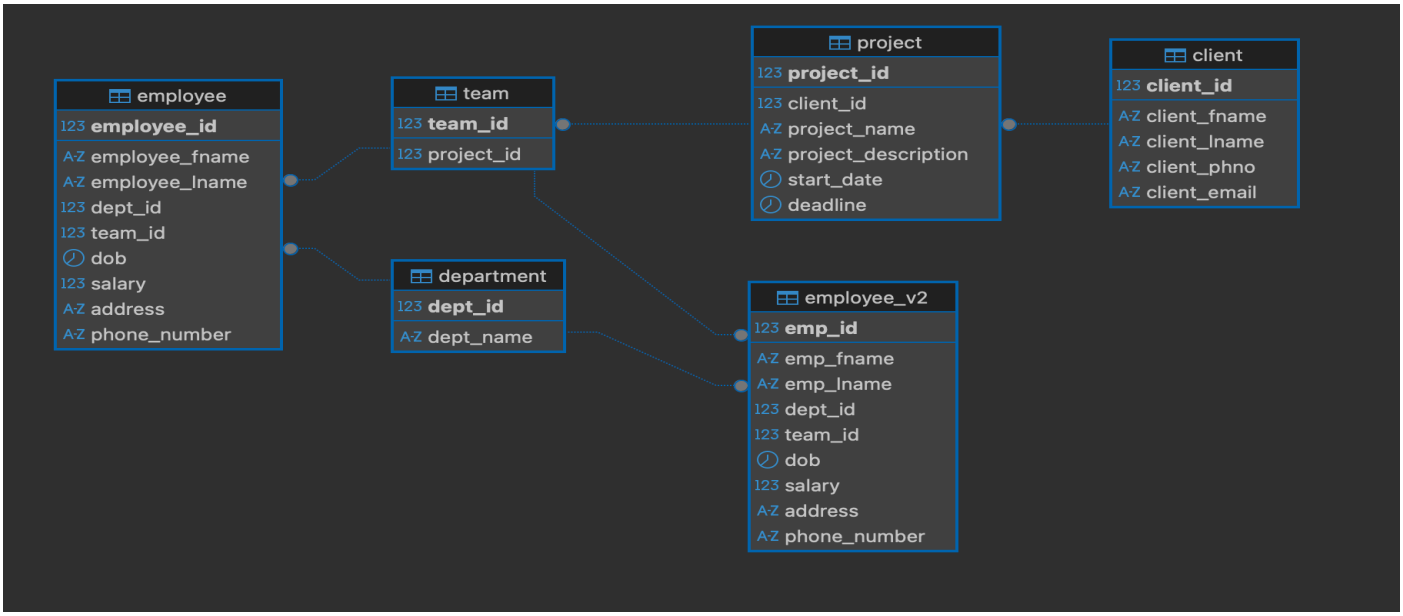
- employees
- departments
- teams
- projects
- clients

within an organization.

However, the current database contains data redundancy and inconsistency issues due to improper handling of relationships among entities.

The database reflects real-world organizational data but does not fully enforce relational integrity, resulting in anomalies such as

- duplicate employee records
- improper department assignments
- ambiguous team–project mappings



No.	Entities Involved	Implemented Relationship (Database)	Explanation
1	Employee ↔ Department	Many-to-One (N:1)	The real-world scenario allows employees to work in multiple departments, but the redesigned `employee_v2` table intentionally restricts each employee to exactly one department to eliminate redundancy and ensure data consistency.
2	Employee ↔ Team	Many-to-One (N:1)	Employees may work in multiple teams in real life, but the current schema enforces a single team per employee to simplify the model and avoid duplication.
3	Team ↔ Project	Many-to-One (N:1)	Multiple teams can work on a single project and teams can work on multiple projects, but the current structure does not support this relationship correctly due to the absence of a junction table.
4	Client ↔ Project	One-to-Many (1:N)	One client can own multiple projects, and each project is correctly linked to a single client using `client_id` as a foreign key.

Queries performed:

1. I observed the execution time before and after indexing the emp_fname column and it actually gave the result with 0.236ms less time!

```
EXPLAIN ANALYZE
SELECT emp_id
FROM employee_v2
WHERE emp_fname = 'Pranav';
```

```
QUERY PLAN
Seq Scan on employee_v2  (cost=0.00..59.00 rows=23 width=4) (actual time=0.015..0.306 rows=23 loops=1)
  Filter: ((emp_fname)::text = 'Pranav'::text)
  Rows Removed by Filter: 1977
Planning Time: 1.106 ms
Execution Time: 0.324 ms
```

```
CREATE INDEX idx_employee_first_name ON employee_v2(emp_fname);
```

```
QUERY PLAN    Ctrl+click to open SQL console
Bitmap Heap Scan on employee_v2  (cost=4.46..37.45 rows=23 width=4) (actual time=0.042..0.066 rows=23 loops=1)
  Recheck Cond: ((emp_fname)::text = 'Pranav'::text)
  Heap Blocks: exact=19
  -> Bitmap Index Scan on idx_employee_first_name  (cost=0.00..4.45 rows=23 width=0) (actual time=0.034..0.034 rows=23 loops=1)
        Index Cond: ((emp_fname)::text = 'Pranav'::text)
Planning Time: 0.494 ms
Execution Time: 0.088 ms
```

Employee Queries

1. **List all employees with their department and team**

```
SELECT e.emp_id, e.emp_fname, e.emp_lname, d.dept_name, t.team_id
FROM employee_v2 e
JOIN department d ON e.dept_id = d.dept_id
JOIN team t ON e.team_id = t.team_id;
```

Shows all employees along with their department and team info.

	emp_id	emp_fname	emp_lname	dept_name	team_id
1	50,001	Akash	Marathe	Information Technology	40,204
2	50,002	Sonali	Khanna	Learning & Development HQ	40,665
3	50,003	Neha	Nair	Customer Support Global	40,540
4	50,004	Sahil	Doshi	Operations HQ	40,614
5	50,005	Mahesh	Doshi	Human Resources India	40,049
6	50,006	Anirudh	Shenoy	Internal Audit Central	40,009
7	50,007	Sonali	Nair	UI/UX Design Central	40,188
8	50,008	Swati	Shenoy	Information Technology	40,040
9	50,009	Varun	Malhotra	Marketing HQ	40,694
10	50,010	Namrata	Reddy	Operations HQ	40,326
11	50,011	Akash	Bhardwaj	Finance	40,434
12	50,012	[NULL]	Gupta	Research & Development – Digital	40,208
13	50,013	Tanvi	Shenoy	Engineering Central	40,113
14	50,014	Monika	Patel	Facilities – Digital	40,038
15	50,015	Ramesh	Verma	Cloud Services – Digital	40,663

2. **Count employees in each department**

```
SELECT d.dept_name, COUNT(*) AS employee_count
FROM employee_v2 e
JOIN department d ON e.dept_id = d.dept_id
GROUP BY d.dept_name
ORDER BY employee_count DESC;
```

Displays number of employees in each department.

	dept_name	employee_count
1	Administration	101
2	Operations HQ	94
3	Supply Chain – Core	93
4	Research & Development – Digital	91
5	Legal Central	89
6	Business Development Central	89
7	Data Science Central	86
8	Facilities – Digital	85
9	Corporate Strategy India	85
10	Cloud Services – Digital	84
11	Procurement India	82
12	Human Resources India	80

3. **Average salary per department**

```
SELECT d.dept_name, ROUND(AVG(e.salary),2) AS avg_salary
FROM employee_v2 e
JOIN department d ON e.dept_id = d.dept_id
GROUP BY d.dept_name;
```

Shows the average salary for employees in each department.

SELECT d.dept_name, ROUND(AVG(e.salary),2) AS

	dept_name	avg_salary
1	Finance	1,421,825.29
2	Customer Support Global	1,470,500.04
3	Cyber Security India	1,470,092.92
4	Data Science Central	1,461,897.59
5	UI/UX Design Central	1,427,963.52
6	Cloud Services – Digital	1,495,195.6
7	DevOps HQ	1,420,803.6
8	Procurement India	1,514,833.8
9	Supply Chain – Core	1,479,099.96
10	Human Resources India	1,428,205.69
11	Facilities – Digital	1,569,721.02
12	Administration	1,481,751.76

4. **Employees with missing info (NULLs)**

```
SELECT count(emp_id)
FROM employee_v2
```

WHERE emp_fname IS NULL OR emp_lname IS NULL OR dob IS NULL OR address IS NULL;

Finds employees with incomplete personal information.

	123 count
1	489

5. **Employees older than 50**

SELECT emp_id, emp_fname, emp_lname, dob
FROM employee_v2
WHERE dob IS NOT NULL AND dob <= CURRENT_DATE - INTERVAL '50 years';

Shows employees who are above 50 years old.

employee_v2 1

SELECT emp_id, emp_fname, emp_lname, dob FRO | Enter a SQL expression to filter resul

	emp_id	emp_fname	emp_lname	dob
1	50,002	Sonali	Khanna	1975-09-16
2	50,007	Sonali	Nair	1965-09-11
3	50,009	Varun	Malhotra	1972-04-16
4	50,011	Akash	Bhardwaj	1975-06-09
5	50,012	[NULL]	Gupta	1975-09-26
6	50,017	Rachana	Bhat	1969-12-18
7	50,025	Ankit	Deshpande	1974-07-31
8	50,027	Namrata	Vora	1972-03-22
9	50,032	Anjali	Sharma	1965-07-21
10	50,033	Daksh	Shah	1965-04-18
11	50,034	Samar	Bansal	1966-04-09
12	50,036	Ritu	Kohli	1965-04-23
13	50,037	Sonal	Kohli	1969-03-25

6. **Top 10 highest-paid employees**

SELECT emp_id, emp_fname, emp_lname, salary
FROM employee_v2
ORDER BY salary DESC

LIMIT 10;

Lists the top 10 employees with the highest salary.

1	51,600	Omkar	Sharma	2,599,806
2	50,818	Ankit	Khanna	2,599,182
3	51,447	Anirudh	Patel	2,598,578
4	50,954	Sonali	Verma	2,597,131
5	51,928	Ramesh	Marathe	2,594,724
6	51,160	Hrithik	Joshi	2,594,486
7	51,153	Aarohi	Kamat	2,593,955
8	51,167	Nisha	Khanna	2,593,570
9	50,102	Jyoti	Doshi	2,592,260
10	50,602	Daksh	Brar	2,590,360

7. **Employees grouped by team**

```
SELECT team_id, COUNT(*) AS total_employees
FROM employee_v2
GROUP BY team_id
ORDER BY total_employees DESC;
```

Shows number of employees in each team.

SELECT team_id, COUNT(*) AS total_employees FR

	team_id	total_employees
1	40,044	9
2	40,599	9
3	40,626	9
4	40,182	8
5	40,345	8
6	40,468	8
7	40,066	8
8	40,151	7
9	40,321	7
10	40,522	7
11	40,180	7
12	40,178	7
13	40,282	7
14	40,545	7
15	40,046	7

8. **Employees without address**

```
SELECT emp_id, emp_fname, emp_lname
FROM employee_v2
WHERE address IS NULL;
```

Lists employees missing their address information.

1	50,003	Neha	Nair
2	50,014	Monika	Patel
3	50,019	Namrata	Kamat
4	50,020	Omkar	Shenoy
5	50,026	Kiran	Verma
6	50,027	Namrata	Vora
7	51,138	Arvind	Kulkarni
8	50,122	Kritika	Deshpande

Department Queries

9. **Departments with more than 80 employees**

```
SELECT d.dept_name, COUNT(*) AS total_employees
FROM employee_v2 e
JOIN department d ON e.dept_id = d.dept_id
GROUP BY d.dept_name
HAVING COUNT(*) > 80;
```

Finds departments that have more than 80 employees.

1	Data Science Central	86
2	Cloud Services – Digital	84
3	Procurement India	82
4	Supply Chain – Core	93
5	Facilities – Digital	85
6	Administration	101
7	Business Development (	89
8	Operations HQ	94
9	Corporate Strategy Indi	85
10	Research & Developmer	91
11	Legal Central	89

10. **Highest paid department**

```
SELECT d.dept_name, SUM(e.salary) AS total_salary
FROM employee_v2 e
JOIN department d ON e.dept_id = d.dept_id
GROUP BY d.dept_name
ORDER BY total_salary DESC
LIMIT 1;
```

Shows which department pays the most overall salary.

1	Administration	149,656,928

11. **Departments with no employees**

```
SELECT d.dept_name
```

```
FROM department d
LEFT JOIN employee_v2 e ON e.dept_id = d.dept_id
WHERE e.emp_id IS NULL;
```

Finds departments that currently have no employees.

dept_name

Team Queries

12. **Teams with the most employees**

```
SELECT team_id, COUNT(*) AS employee_count
FROM employee_v2
GROUP BY team_id
ORDER BY employee_count DESC
LIMIT 5;
```

Top 5 teams with the highest number of employees.

1	40,599	9
2	40,626	9
3	40,044	9
4	40,468	8
5	40,066	8

13. **Teams working on projects**

```
SELECT t.team_id, p.project_name
FROM team t
JOIN project p ON t.project_id = p.project_id
Order by p.project_name;
```

Shows which projects each team is working on.

	team_id	project_name
1	40,570	AI App 337
2	40,441	AI App 337
3	40,677	AI App 439
4	40,296	AI App 820
5	40,563	AI Cloud 649
6	40,678	AI Cloud 654
7	40,226	AI Cloud 911
8	40,336	AI Cloud 911
9	40,589	AI Cloud 911
10	40,610	AI Cloud 962
11	40,084	AI Dashboard 721
12	40,263	AI Dashboard 721
13	40,293	AI Dashboard 968
14	40,596	AI Dashboard 968

14.**Average employee salary per team**

```
SELECT team_id, ROUND(AVG(salary),2) AS avg_salary
FROM employee_v2
GROUP BY team_id
ORDER BY avg_salary DESC;
```

Displays average salary of employees for each team.

	team_id	avg_salary
1	40,491	2,564,657
2	40,623	2,545,793
3	40,356	2,531,045
4	40,213	2,528,094.5
5	40,482	2,502,066
6	40,130	2,464,885
7	40,158	2,451,908

15.**Teams with only one employee**

```
SELECT team_id
FROM employee_v2
GROUP BY team_id
HAVING COUNT(*) = 1;
```

Identifies teams that have only one employee.

1	40,102
2	40,130
3	40,483
4	40,530
5	40,218
6	40,669
7	40,225

There are 115 such teams

Project Queries

16. **Active projects (current date between start_date and end_date)**

```
SELECT project_id, project_name, start_date, deadline
FROM project
WHERE CURRENT_DATE BETWEEN start_date AND deadline;
```

Shows projects that are currently in progress.

1	30,004	EdTech App 404	2025-01-12	2026-11-30
2	30,006	Logistics Portal 312	2025-07-17	2027-12-15
3	30,008	HRMS Portal 291	2025-05-28	2027-07-22
4	30,009	LMS Dashboard 549	2025-07-05	2026-07-07
5	30,010	EdTech Cloud 443	2024-01-05	2026-01-27
6	30,012	IoT Portal 990	2025-08-08	2028-02-04
7	30,019	CyberSecurity Portal 388	2025-11-17	2027-12-20
8	30,022	Inventory Solution 898	2025-11-16	2027-04-24
9	30,029	HealthTech Portal 792	2023-10-16	2026-05-07

17. **Projects with no team assigned**

```
SELECT p.project_id, project_name
FROM project p
LEFT JOIN team t ON p.project_id = t.project_id
WHERE t.team_id IS NULL;
```

Finds projects that currently don’t have any teams assigned.

1	30,462	EdTech Solution 525
2	30,688	FinTech Cloud 845
3	30,044	HealthTech System 788
4	30,417	IoT App 878
5	30,470	Inventory Platform 549
6	30,140	Logistics Cloud 121
7	30,263	Mobile Banking Dashboard 79
8	30,350	Inventory Solution 128
9	30,342	Fleet Cloud 855
10	30,357	Fleet Cloud 388
11	30,234	IoT Mobile 875
12	30,216	ERP Platform 422

18. **Longest running project**

```
SELECT project_id, project_name, deadline - start_date AS duration_days
FROM project
ORDER BY duration_days DESC
LIMIT 1;
```

Shows the project with the longest planned duration.

1	30,242	Retail Cloud 561	1,095

19. **Number of teams per project**

```
SELECT p.project_name, COUNT(t.team_id) AS total_teams
FROM project p
LEFT JOIN team t ON p.project_id = t.project_id
GROUP BY p.project_name
ORDER BY total_teams DESC;
```

Displays how many teams are working on each project.

	project_name	total_teams
1	Analytics Suite 133	6
2	HMS Solution 787	5
3	Fleet Cloud 82	5
4	POS Suite 914	5
5	HealthTech Portal 792	5
6	ERP System 485	4
7	Analytics Platform 619	4
8	Logistics Portal 860	4
9	Fleet App 49	4

Client Queries

20. **Clients with multiple projects**

```
SELECT c.client_fname, c.client_lname, COUNT(p.project_id) AS projects_count
FROM client c
JOIN project p ON c.client_id = p.client_id
GROUP BY c.client_fname, c.client_lname
HAVING COUNT(p.project_id) > 1;
```

Lists clients who have more than one project.

	client_fname	client_lname	projects_count
1	Riya	Verma	2
2	Atharv	Trivedi	2
3	Krishna	Verma	2
4	Myra	Sharma	2
5	Aaradhya	Singh	3
6	Krishna	Patel	2
7	Aarav	Saxena	2
8	Ananya	Yadav	2
9	Ayaan	Desai	2
10	Anika	Kapoor	5
11	Krishna	Joshi	2
12	Ishaan	Dubey	4

21. **Projects by client**

```
SELECT c.client_fname, c.client_lname, p.project_name
FROM client c
JOIN project p ON c.client_id = p.client_id;
```

Shows which projects belong to which client.

1	Riya	Khan	SmartCity Portal 756
2	Diya	Sharma	Inventory Mobile 261
3	Ananya	Ghosh	Mobile Banking Solution 201
4	Rohan	Sharma	EdTech App 404
5	Ananya	Reddy	Fleet Solution 418
6	Vikram	Sharma	Logistics Portal 312
7	Aanya	Patel	AgriTech App 386
8	Ishaan	Chauhan	HRMS Portal 291
9	Riya	Pai	LMS Dashboard 549
10	Reyansh	Gupta	EdTech Cloud 443
11	Myra	Bose	AgriTech Suite 593
12	Reyansh	Shetty	IoT Portal 990

22. **Clients without projects**

```
SELECT c.client_fname, c.client_lname
FROM client c
LEFT JOIN project p ON c.client_id = p.client_id
WHERE p.project_id IS NULL;
```

Finds clients that don’t have any projects assigned.

1	Sara	Desai
2	Shreya	Khan
3	Myra	Pandey
4	Arjun	Naik
5	Shreya	Pandey
6	Karan	Kapoor
7	Aaradhya	Chauhan
8	Aditya	Mishra
9	Nikhil	Mishra

Salary & Performance Queries

23. **Employees earning below average salary**

```
SELECT emp_id, emp_fname, emp_lname, salary
FROM employee_v2
WHERE salary < (SELECT AVG(salary) FROM employee_v2);
```

Finds employees whose salary is below the company average.

	emp_id	emp_fname	emp_lname	salary
1	50,001	Akash	Marathe	436,347
2	50,003	Neha	Nair	1,098,655
3	50,005	Mahesh	Doshi	645,851
4	50,006	Anirudh	Shenoy	401,914
5	50,007	Sonali	Nair	1,198,135
6	50,009	Varun	Malhotra	1,339,775
7	50,010	Namrata	Reddy	1,339,169
8	50,011	Akash	Bhardwaj	856,983

	avg
1	1,469,854.55

24. **Total salary per department**

```
SELECT d.dept_name, SUM(e.salary) AS total_salary
FROM employee_v2 e
JOIN department d ON e.dept_id = d.dept_id
GROUP BY d.dept_name
ORDER BY total_salary DESC;
```

Shows the total salary expense per department.

	dept_name	total_salary
1	Administration	149,6
2	Operations HQ	143,9
3	Supply Chain – Core	137,556,296
4	Research & Development – Digital	136,864,366
5	Facilities – Digital	133,426,287
6	Business Development Central	129,234,016
7	Legal Central	129,153,239
8	Data Science Central	125,723,193

25.**Top 5 youngest employees**

```
SELECT emp_id, emp_fname, emp_lname, dob
FROM employee_v2
WHERE dob IS NOT NULL
ORDER BY dob DESC
LIMIT 5;
```

Displays the 5 youngest employees in the company.

1	50,350	Mohit	Kulkarni	2004-12-29
2	50,167	Sonal	Kamat	2004-12-28
3	51,010	Anjali	Joshi	2004-12-26
4	50,590	Nikhil	Khanna	2004-12-10
5	50,577	Sonal	Deshpande	2004-12-10